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# --- Setup (uses installed TF >=2.18)
import tensorflow as tf, numpy as np, random, os
from tensorflow import keras
from tensorflow.keras import layers
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, classification_report

print("TensorFlow version:", tf.__version__)
np.random.seed(42); random.seed(42); tf.random.set_seed(42)

# --- Load CIFAR-10
(x_train, y_train), (x_test, y_test) = keras.datasets.cifar10.load_data()
y_train = y_train.flatten(); y_test = y_test.flatten()
num_classes = 10

# --- Preprocess
x_train = x_train.astype("float32") / 255.0
x_test = x_test.astype("float32") / 255.0

# Validation split
val_ratio = 0.1
n_val = int(len(x_train) * val_ratio)
x_val, y_val = x_train[:n_val], y_train[:n_val]
x_train, y_train = x_train[n_val:], y_train[n_val:]

# --- Data augmentation
data_augmentation = keras.Sequential([
    layers.RandomFlip("horizontal"),
    layers.RandomRotation(0.05),
    layers.RandomZoom(0.1),
])

# --- Model
def make_cnn(input_shape=(32,32,3), num_classes=10):
    inputs = keras.Input(shape=input_shape)
    x = data_augmentation(inputs)
    x = layers.Conv2D(32, 3, padding="same", activation="relu")(x)
    x = layers.BatchNormalization()(x)
    x = layers.MaxPooling2D()(x)

    x = layers.Conv2D(64, 3, padding="same", activation="relu")(x)
    x = layers.BatchNormalization()(x)
    x = layers.MaxPooling2D()(x)

    x = layers.Conv2D(128, 3, padding="same", activation="relu")(x)
    x = layers.BatchNormalization()(x)
    x = layers.MaxPooling2D()(x)

    x = layers.Dropout(0.3)(x)
    x = layers.Flatten()(x)
    x = layers.Dense(256, activation="relu")(x)
    x = layers.Dropout(0.4)(x)
    outputs = layers.Dense(num_classes, activation="softmax")(x)
    return keras.Model(inputs, outputs, name="cnn_classifier")

model = make_cnn()
model.compile(optimizer=keras.optimizers.Adam(1e-3),
              loss="sparse_categorical_crossentropy",
              metrics=["accuracy"])
model.summary()

# --- Train
callbacks = [
    keras.callbacks.EarlyStopping(patience=5, restore_best_weights=True),
    keras.callbacks.ReduceLROnPlateau(patience=2, factor=0.5)
]
history = model.fit(
    x_train, y_train,
    validation_data=(x_val, y_val),
    epochs=30, batch_size=64, callbacks=callbacks, verbose=2
)

# --- Evaluate
test_loss, test_acc = model.evaluate(x_test, y_test, verbose=0)
```

```
print("Test accuracy: {test_acc:.4f} ")

# --- Report + Confusion Matrix
y_pred = model.predict(x_test, verbose=0).argmax(axis=1)
cifar10_names = ["airplane", "automobile", "bird", "cat", "deer",
                  "dog", "frog", "horse", "ship", "truck"]
print(classification_report(y_test, y_pred, target_names=cifar10_names))
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6,6))
plt.imshow(cm, interpolation='nearest')
plt.title("Confusion Matrix")
plt.colorbar()
plt.xlabel("Predicted")
plt.ylabel("True")
plt.tight_layout()
plt.show()

# --- Save (Keras 3 format)
model.save("cnn_model.keras")
np.savez("cnn_metrics_cifar10.npz", test_acc=test_acc)
```


TensorFlow version: 2.19.0
Model: "cnn_classifier"

Layer (type)	Output Shape	Param #
input_layer_2 (InputLayer)	(None, 32, 32, 3)	0
sequential_1 (Sequential)	(None, 32, 32, 3)	0

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batch_normalization_3 (BatchNormalization)	(None, 32, 32, 32)	128
max_pooling2d_3 (MaxPooling2D)	(None, 16, 16, 32)	0
conv2d_4 (Conv2D)	(None, 16, 16, 64)	18,496
batch_normalization_4 (BatchNormalization)	(None, 16, 16, 64)	256
max_pooling2d_4 (MaxPooling2D)	(None, 8, 8, 64)	0
conv2d_5 (Conv2D)	(None, 8, 8, 128)	73,856
batch_normalization_5 (BatchNormalization)	(None, 8, 8, 128)	512
max_pooling2d_5 (MaxPooling2D)	(None, 4, 4, 128)	0
dropout_2 (Dropout)	(None, 4, 4, 128)	0
flatten_1 (Flatten)	(None, 2048)	0
dense_2 (Dense)	(None, 256)	524,544

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